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ACADEMIC CREDENTIALS

B.Sc. in Electrical & Electronic Engineering
MYMENSINGH ENGINEERING COLLEGE, UNIVERSITY OF DHAKA
CGPA: 3.27/4.00

28 August 2025
Mymensingh, Bangladesh

Current Research Interests

- **mmWave/Microwave Antennas & RF Hardware:** Design and beamforming of high-directivity microstrip patch arrays for 5G/6G and radar, coupled with RF front-end PCB design, impedance matching, and VNA-based characterization.
- **Metamaterials & THz Biosensors:** Engineering functional metasurfaces and metamaterial absorbers for microwave RCS reduction, device performance enhancement, and high-sensitivity, label-free biomolecular/refractive index sensing.
- **Computational Electromagnetics & ML-Assisted Sensing:** Full-wave numerical modeling (CST Studio Suite) across THz and microwave regimes, integrated with explainable machine learning models for rapid parameter extraction and bio-detection.

PROFESSIONAL EXPERIENCES

Embedded System Engineer | DhumketuX Defense, Mymensingh, Bangladesh | Feb 2023 – Feb 2024
[DhumketuX – First Rocket Company of Bangladesh](#)

- Designed, simulated, and optimized a 433 MHz high-gain Yagi-Uda antenna (~10 dB gain) using CST Studio Suite for telemetry and RF communication systems.
- Performed full-wave electromagnetic simulations and antenna optimization focusing on impedance matching, gain, and radiation characteristics.
- Participated in antenna prototyping, fabrication, and validation using Vector Network Analyzer (VNA) measurements.
- Worked with STM32, telemetry modules, GPS, IMU, ESCs, and flight controllers for aerospace and UAV applications.
- Contributed to avionics integration, FPV (5.8 GHz) communication systems, and IoT-based wireless rocket ignition systems.
- Conducted 15+ rocket engine test campaigns involving instrumentation, data acquisition, performance evaluation, and failure analysis.

Research Analyst

Softsasi NextGen Research, Mymensingh, Bangladesh | Dec 2025 – Sep 2026
<https://research.softsasi.com/>

- Conducted research on metamaterials, THz biosensing, and computational electromagnetics for biomedical sensing applications.
- Performed CST Studio Suite and SCAPS-1D simulations to develop THz biosensors and lead-free perovskite solar cells, resulting in peer-reviewed publications in Elsevier and Springer Nature journals.

PUBLICATIONS

Journal Articles

- *Explainable Machine Learning Assisted Terahertz Metamaterial Absorber for Multi-Cancer Cell Bio-Sensing.* Next Chemical Engineering, Vol. 3, Article 100124, Elsevier, 2026.
- *Numerical Investigation of Optoelectronic Performance Limits in Lead-Free $(\text{FA})_{0.5}(\text{MA})_{0.5}\text{SnI}_3$ Perovskite Solar Cells Using SCAPS-1D.* Optical and Quantum Electronics (Springer Nature), 2026.

International Conference Papers

- *Design and Optimization of a Thin, Multi-Resonant Metamaterial Absorber for Enhanced Electromagnetic Absorption in C and X Bands,* 2025 International Conference on (ECCE), Chittagong, Bangladesh, 2025.
- *Design and Optimization of Bacteriophage Virus Shape Tuneable Microwave Metamaterial Absorber for S, C band Applications.* 27th International Conference on Computer and Information Technology (ICCIT), 2024.
- *A Machine Learning Approach for Accurate RMSD Estimation.* Accepted and presented in 2nd International Conference on Next-Generation Computing, IoT and Machine Learning, Bangladesh, 2025.
- *Improving PCE of Lead Free, Organic, Inorganic, Metal Halide Bilayer Perovskite Solar Cells through Optimization of Bandgap, Defect Density and Electron Affinity with Different ETLs,* Dhaka, Bangladesh, 2025 .

PROJECTS & PRESENTATION

- Oral Presenter, “Bacteriophage Virus-Shaped Metamaterial Absorber for S & C Bands” — ICCIT 2024
- 77 GHz mmWave 1×4 Microstrip Patch Antenna Array — Designed and optimized in CST Studio Suite for radar applications, achieving 12.39 dBi directivity and 89% radiation efficiency.
- 433 MHz High-Gain Yagi-Uda Antenna — Designed, simulated, fabricated, and tested for telemetry and RF communication systems, achieving approximately 10 dB gain.
- Dual-Band Microstrip Patch Antenna Array (1.5 GHz GPS / 2.4 GHz WLAN) — Optimized for enhanced impedance matching, gain, and radiation performance.
- Custom STM32WB55 BLE Module (4-Layer PCB) — Designed in KiCad 10 with RF layout optimization, USB Type-C, crystal oscillators, and wireless communication interfaces.
- Custom STM32 Development Board (4-Layer PCB) — Designed in Altium Designer integrating USB, SWD, power regulation, and peripheral interfaces.
- Explainable ML Framework for THz Biosensing — Developed an explainable machine learning (XAI) framework coupled with CST full-wave simulations to predict resonance frequency shifts and classify cancer cell refractive indices with high sensitivity and minimal computation time.

TECHNICAL SKILLS

- **RF, EM & Antenna Engineering:** CST Studio Suite, Ansys HFSS, ADS, Lumerical, SCAPS-1D, MATLAB, Patch Antenna Arrays, mmWave, THz Sensing, Metamaterial Absorbers.
- **Hardware, Embedded & Programming:** Altium Designer, KiCad (4-Layer PCB), STM32, Arduino, Proteus, C Programming, Applied Machine Learning (Supervised Learning, Regression/Classification for EM data, Explainable AI / SHAP)
- **Research, Measurement & Analysis:** Vector Network Analyzer (VNA), Spectrum Analyzer, Oscilloscope, OriginPro, Adobe Illustrator, EndNote, Mendeley, Power BI.

WORKSHOP EXPERIENCES & CERTIFICATIONS

Training on Industrial Technology in Electrical Engineering & Instrumentation | Apr 2025- May 2025

Training Institute for Chemical Industries (TICI), BCIC, Polash, Narsingdi, Bangladesh

- Completed hands-on training in Electrical Testing & Measurement, Industrial Instrumentation, Sensors, Control Systems, Hardware Integration, and Electrical Safety Engineering.
- Gained practical exposure to Oscilloscope, Spectrum Analyzer, VNA, and Digital Multimeter for electrical/RF measurements, troubleshooting, and performance evaluation. Final Grade: A (75-<80%)

AWARDS & EXTRACURRICULAR ACTIVITIES

- **First Runner-up, Walton laptop presents 3rd national science fest 2023,** (BAU, Bangladesh)
- **Second runner-up, NASA Space Apps Challenge-2024, Dhaka.**
- Member and Researcher at Alpha Science Lab.
- Executive at BASIS Student's Forum, MEC Chapter , Mymensingh , Bangladesh
- Volunteer at Devsphere Hackathon 2025, Mymensingh Engineering College.
- Member and Editor at Mymensingh Engineering College (MEC) Research Community.
- Campus Ambassador , ICT Olympiad Bangladesh, Dhaka, Bangladesh | Mar 2022 – 2025

REFERENCE

S.M. Anowarul Haque Associate Professor & Head, Department of Electrical and Electronic Engineering Mymensingh Engineering College (University of Dhaka) Mymensingh-2208, Bangladesh Email: anowar.eee@mec.ac.bd Phone no: +8801816419622	Abdul Wahed Lecturer, Department of Electrical and Electronic Engineering Mymensingh Engineering College, (University of Dhaka) Mymensingh-2208, Bangladesh. Email: wahed.eee8421@mec.ac.bd Phone no: +8801721343362
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